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HYDRAU LAB 2 – PET 1 – GROUP 2	OPEN CHANNEL	RATING:

I. OBJECTIVE

To know the effect of channel slope and hydraulic radius to the friction factor (f).

II. APPARATUS AND MATERIALS

1. Hydraulic Laboratory Setup
2. Stopwatch
3. Measuring Devices (Ruler, Caliper)
4. Graphing Paper

III. PROCEDURE

1. Set an initial inclination of the channel (Refer to the protractor at the far end of the channel).
2. Fit the 1-1/6" diameter orifice to the bourdon mouthpiece of the reservoir (tank)
3. Fill the reservoir with water by pumping from the lower reservoir until it reaches a stationary level.
4. Determine the flow by weighing the collected amount of water in the weigh tank and timing it. (Follow procedures in exercise D-1) Countercheck the flow rate using the orifice (Exercise D-4).
5. Measure the weight and normal depth of water flowing in the channel and record it.
6. Record the reading of the protractor, indicating the inclination of the channel. Use the relation; $S_o = \tan(\theta)$
7. Compute for the roughness coefficient (n) using Manning's equation.

$$n = \frac{A(Rh)^{\frac{2}{3}}S_o^{\frac{1}{2}}}{Q}$$

Where:

n = Roughness Coefficient

A = Cross-sectional area of channel (m²)
(Normal depth x width of channel)

RH= Hydraulic radius (m)

(Cross-sectional/weighted perimeter: A/P)

P = 2(Normal depth) + width of channel

S_o = Channel slope

Q = Flow rate (m³ / s)

8. Compute for the frictional coefficient (f)

$$f = \frac{78.48(n)^2}{Rh^{\frac{1}{3}}} = \frac{19.62n^2}{R^{\frac{1}{3}}}$$

9. Vary the inclination of the channel and repeat procedure (4) to (9).
 10. Make at least six variations.
 11. Construct an individual graphs, of frictional coefficient (f vs. slope of channel (So) and frictional coefficient (f) vs. hydraulic radius (Rh).

IV. DATA AND TABULATION

A. Rectangular Contracted Weir

	TRIAL 1	TRIAL 2
SLOPE	0.03273	0.03273
VOLUME OF WATER (mL)	2530 mL	2845 mL
TIME ELAPSED (sec)	8.23s	9.43s
FLOW RATE (cm ³ / sec)	307.4119 (cm ³ / sec)	301.6967 (cm ³ / sec)
NORMAL DEPTH (cm)	1 cm	1 cm
WIDTH OF CHANNEL (cm)	5.2 cm	5.2 cm
WETTED AREA (cm ²)	5.2 cm	5.2 cm
WETTED PERIMETER (cm)	7.2 cm	7.2 cm
HYDRAULIC RADIUS (cm)	0.7222 cm	0.7222 cm
n	0.002463	0.0029100
f	0.00013266	0.00013777

V. COMPUTATION

Trial 1 Calculations

1. Hydraulic Radius (R_h)

$$R_h = \frac{5.2}{7.2} = \boxed{0.7222 \text{ cm}}$$

2. Volumetric Flow Rate (Q_1)

$$Q_1 = \frac{2530}{8.23} = \boxed{307.4119 \text{ cm}^3/\text{s}}$$

3. Manning's Roughness Coefficient (n_1)

$$n_1 = \frac{R_h^{\frac{2}{3}} \cdot \sqrt{s}}{Q_1} = \frac{(0.7222)^{\frac{2}{3}} \cdot \sqrt{0.03273}}{307.4119} = \boxed{0.002463}$$

4. Darcy-Weisbach Friction Factor (f_1)

$$f_1 = \frac{19.62n_1^2}{R_h^{\frac{1}{3}}} = \frac{19.62(0.002463)^2}{(0.7222)^{\frac{1}{3}}} = \boxed{0.00013266}$$

Trial 2 Calculations

1. Hydraulic Radius (R_h)

$$R_h = \frac{5.2}{7.2} = \boxed{0.7222 \text{ cm}}$$

2. Volumetric Flow Rate (Q_2)

$$Q_2 = \frac{2845}{9.43} = \boxed{301.6967 \text{ cm}^3/\text{s}}$$

3. Manning's Roughness Coefficient (n_2)

$$n_2 = \frac{R_h^{\frac{2}{3}} \cdot \sqrt{s}}{Q_2} = \frac{(0.7222)^{\frac{2}{3}} \cdot \sqrt{0.03273}}{301.6967} = \boxed{0.002510}$$

4. Darcy-Weisbach Friction Factor (f_2)

$$f_2 = \frac{19.62n_2^2}{R_h^{\frac{1}{3}}} = \frac{19.62(0.002510)^2}{(0.7222)^{\frac{1}{3}}} = \boxed{0.00013777}$$

Average Values

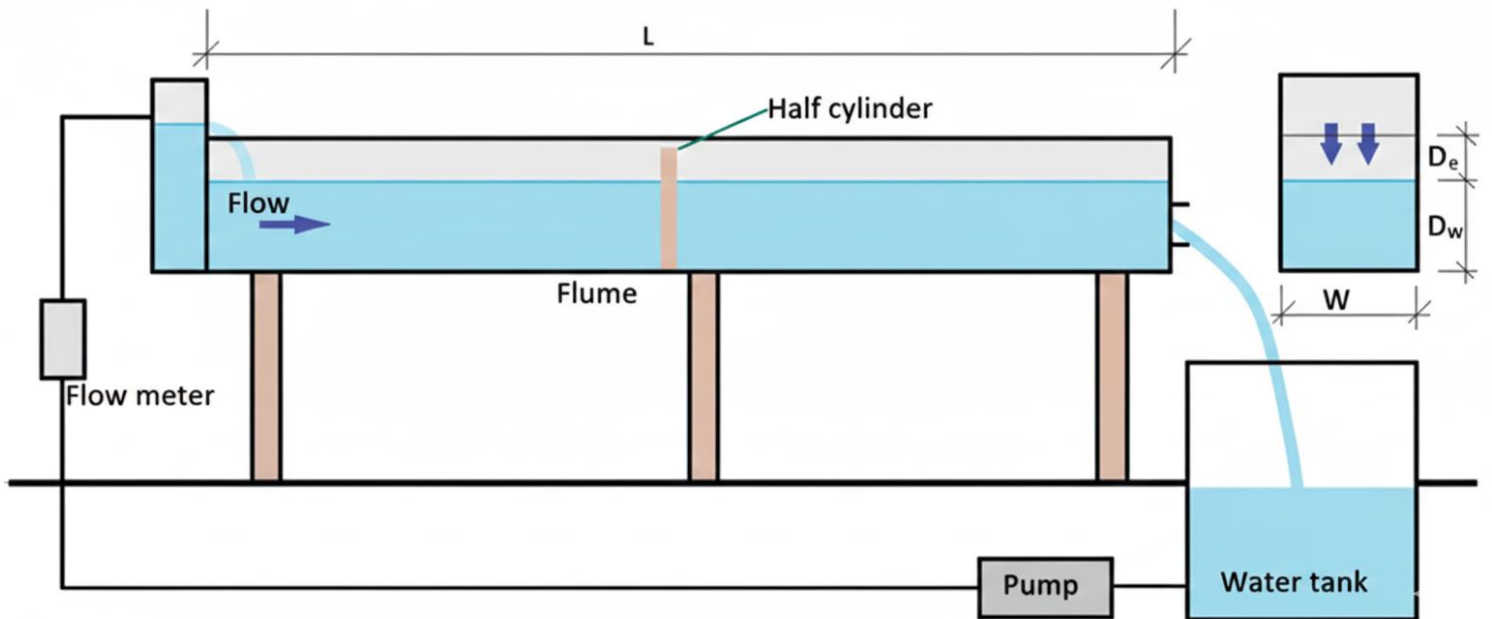
1. Average Manning's Coefficient (n_{ave})

$$n_{ave} = \frac{n_1 + n_2}{2} = \frac{0.002463 + 0.002510}{2} = \boxed{0.002487}$$

2. Average Friction Factor (f_{ave})

$$f_{ave} = \frac{f_1 + f_2}{2} = \frac{0.00013266 + 0.00013777}{2} = \boxed{0.0001352}$$

VI. FIGURE



VII. ANALYSIS AND INTERPRETATION

The Manning's equation and the Darcy-Weisbach equation are two fundamental models used to quantify frictional energy losses in open channel flow, and while distinct in their approach, they are intrinsically linked. The **Manning's equation** is characterized by the **Manning's roughness coefficient (n)**, an empirical measure reflecting the physical characteristics of the channel boundary, such as surface roughness and irregularity. It is most frequently used in civil engineering for practical design and is derived from a velocity perspective, where a higher n signifies greater flow resistance. In contrast, the **Darcy-Weisbach equation** is defined by the dimensionless **friction factor (f)**, which quantifies the energy head loss directly and is more fundamentally rooted in fluid mechanics, accounting for both flow characteristics (like the Reynolds number) and relative roughness. The relationship $f = \frac{8gn^2}{R_h^{1/3}}$ mathematically confirms their connection, demonstrating that both coefficients describe the same physical reality of frictional resistance. The experimental results, with an average roughness $n_{ave} = \mathbf{0.002487}$ and an average friction factor $f_{ave} = \mathbf{0.0001352}$, are consistent with this interrelation, both indicating that the channel possesses an exceptionally smooth internal surface, leading to minimal resistance to fluid flow.

VIII. DOCUMENTATION

